Homework #3: Vector Embeddings and RAG

Write-Up Report

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Assignment: Homework #3 - Vector Embeddings and RAG

Collaborators (coding and report): None. GenAI disclosure: Claude (Anthropic) was used as a coding assistant, to help with the manual scoring, and to draft this report.

Files submitted: `hw3_embeddings_AsadKhan.ipynb` (completed starter notebook, renamed) and this report.

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1. Introduction and Methodology

1.1 RAG and the goal of the experiment

Retrieval-Augmented Generation (RAG) answers a question in two steps. A **retriever** embeds the question and returns the k text chunks whose embeddings are closest to it in a vector store. A **generator** (here gpt-4o-mini) is then asked to answer using only those chunks. The LLM can only be as good as the evidence it is handed: if the retriever never surfaces a passage, the generator cannot use it, however capable it is. The goal of this homework was to measure that dependency on a realistic corpus. Experiment A compares two sentence-transformer embedders, `all-MiniLM-L6-v2` (384 dimensions, 6 layers) and `all-mpnet-base-v2` (768 dimensions, 12 layers). Experiment B varies chunk size, chunk overlap and k for MiniLM. Five questions of increasing difficulty were scored by hand.

1.2 Corpus

Table 1. The four documents after loading and chunking (default chunk_size 1000, overlap 200).

Short name	Document	Pages	Characters	Chunks
Blueprint	White House Blueprint for an AI Bill of Rights (2022)	73	220,968	299
EO 14110	Executive Order 14110 on Safe, Secure, and Trustworthy AI (2023), Federal Register print	36	145,506	194
NIST AI RMF	NIST AI 100-1, AI Risk Management Framework 1.0	48	106,444	148
EU AI Act	EU Artificial Intelligence Act, COM(2021) 206 proposal	108	314,714	424
Total		**265**	**787,632**	**1,065**

Inspecting the files showed something the assignment description does not mention. The "EU AI Act" file is not a short summary of key principles: it is the full 108-page 2021 Commission proposal (explanatory memorandum, 89 recitals, 85 articles and the financial statement), *without* its annexes. It alone supplies 40% of all chunks, so with a fixed k it has the most chances to crowd out the smaller

documents. Its legal register ("this Regulation", "the Union", "providers") also differs sharply from the US documents, which turned out to matter for retrieval.

1.3 Pipeline and text-splitting strategy

1. **Loading (TODO 1).** `PyPDFLoader` yields one `Document` per page. Each page is tagged with metadata (`source_title`, a short name such as "EU AI Act", and a 1-based `page_number`), so every chunk can be traced back to its document and page. The loader uses the local `data/` copy when present and otherwise downloads from the course GitHub, so the notebook also runs on Colab.
2. **Cleaning.** PDF extraction leaves artifacts that waste embedding capacity. I removed the Federal Register running header that repeats on every EO page, the Blueprint's "SECTION TITLE" stamps and the EU "EN n EN" footers. I re-joined words hyphenated across line breaks ("respon-sible") and collapsed visual line breaks into running text. Some artifacts cannot be repaired by regex. The most important is that superscripts are flattened, so 10^{26} appears in the text as "10 26". I left these in deliberately, as a test of the generator.
3. **Splitting (TODO 2).** `RecursiveCharacterTextSplitter` with chunk_size 1000, overlap 200 and the separator list `["\n\n", "\n", ". ", "; ", ", ", " ", ""]` with `keep_separator="end"`. Once line breaks are collapsed, the splitter cuts at sentence ends first, then clause boundaries, and only splits inside a sentence as a last resort. `"end"` keeps the period with its sentence instead of starting the next chunk with ". ". Chunks shorter than 100 characters (cover pages, page-number fragments) are dropped. Mean chunk length is 768 characters, about 159 word-piece tokens.
4. **Embedding and indexing (TODO 3).** `HuggingFaceEmbeddings` with `normalize_embeddings=True` and a Chroma collection using cosine distance (`hnsw:space = cosine`). Each model gets its own persistent database, and the collection is deleted before it is rebuilt, so re-running the notebook never inserts duplicates.
5. **Generation (TODO 5).** The top-k chunks are formatted as labeled excerpts ("[Excerpt 2] (EO 14110, p. 7)") and sent through a LangChain chain (`PromptTemplate | ChatOpenAI | StrOutputParser`) with temperature 0. The prompt tells the model to use only the context, cite the excerpt behind each claim, treat each document separately and then synthesize, and say explicitly when a document or fact is missing. Answers are cached on disk keyed by a hash of (model, context, question), so re-running the notebook reproduces exactly the answers I scored.

Rationale for a multi-document corpus. I split per page rather than per document. The cost is that a chunk can never span a page break. The benefit is that every chunk carries an exact page citation, which the generator uses and which made manual verification fast. 1000 characters, about one legal provision or one paragraph of prose, is long enough to hold a complete rule ("both of the following conditions are fulfilled: (a) ... (b) ..."). It is also short enough that its embedding is not an average of several unrelated topics, and well under MiniLM's 256-token input limit (only 3.2% of default chunks exceed it). The 200-character overlap protects a sentence that straddles a cut. I did *not* prepend the document title to each chunk's text. That would help any query that names a document, but it would

also hide exactly the difference between embedders that Experiment A is meant to measure. Section 4 returns to this choice.

1.4 Queries (TODO 4)

I wrote the questions myself after reading the relevant sections, aiming for questions that a single lucky chunk cannot answer. Each has several required parts, and each part is located in a known place, so retrieval can be checked chunk by chunk.

Table 2. The five test queries.

#	Query	Type, location of the answer and why it is hard
Q1	*Under the Blueprint for an AI Bill of Rights' 'Human Alternatives, Consideration, and Fallback' principle, when should people be able to opt out of an automated system, what must a fallback and escalation process provide, and which sensitive domains call for extra human oversight?*	Single document (Blueprint, pp. 7, 46-49). Three parts. The sensitive-domain list (criminal justice, employment, education, health) sits at the *end* of the principle statement, after the opt-out and fallback text.
Q2	*Under Executive Order 14110, what must companies developing dual-use foundation models report to the federal government, and what computing-power thresholds trigger these reporting requirements for models and for computing clusters?*	Single, different document (EO Sec. 4.2, p. 7). Needs the reporting list (A)-(C) and three numeric thresholds (10^{26} operations; 10^{23} for biological sequence data; clusters with 100 Gbit/s networking and 10^{20} operations per second), which appear in the text as "10 26".
Q3	*What transparency and explanation duties do the Blueprint for an AI Bill of Rights, the EU AI Act, and the NIST AI RMF each place on those who build or deploy AI systems, and what common understanding of why transparency matters emerges across them?*	Multi-document synthesis. Search and combine: find the transparency duty in three documents (Blueprint "Notice and Explanation" principle, p. 6 and pp. 40-45; EU Art. 13 "Transparency and provision of information to users", p. 51, and Art. 52 "Transparency obligations", p. 70; NIST Sec. 3.4-3.5 "Accountable and Transparent", "Explainable and Interpretable", pp. 20-21). Thematic synthesis: state why all three think transparency matters. In the extracted Blueprint text, "notice" and "explanation" occur 55 and 60 times, against 22 for "transparen-".
Q4	*How does the NIST AI RMF's voluntary approach to managing AI risk differ from the EU AI Act's legally binding obligations for high-risk AI systems, in terms of who must comply, how risk is assessed, and what happens before a system can be deployed?*	Comparative, with three axes of contrast. Needs NIST's "voluntary, non-prescriptive" design and EU's ex-ante conformity assessment, registration and provider obligations.
Q5	*How does the EU AI Act proposal define an 'AI system', and under Article 6, what conditions make an AI system 'high-risk'?*	Definition and scope inside the longest document. Two exact passages about 6 pages apart: Art. 3(1) on p. 40 and Art. 6(1)-(2) on p. 46. The explanatory memorandum and recitals paraphrase both passages and compete with the real text.

A pilot finding. My first Q5 asked which "techniques and approaches listed in Annex I" fall under the AI-system definition. Every run failed to retrieve Annex I. Searching the extracted text showed why: the annexes are a separate EU document and are not in this PDF. Both models' answers stated that the Annex I list was not in the context rather than inventing one (mpnet even retrieved the Art. 3(1) definition, which points to Annex I). This was a good sign for the prompt's honesty rule, but an

unanswerable question cannot measure retrieval quality, so I replaced it with the Article 6 question above.

1.5 Evaluation

The starter's `evaluate_results` prompts with `input()`. I kept that path (set `INTERACTIVE_EVAL = True`) but recorded the 55 judgments (11 configurations x 5 queries) in a `MANUAL_SCORES` table, so the notebook runs end to end and shows every chunk and score. Every score was assigned after reading every retrieved chunk in full and checking the answer against the source text. I applied the rubric as follows:

- **Retrieval:** 3 = the chunks are on-topic, from the right document(s), and contain the evidence for every part of the question; 2 = relevant material but a required part or a required document is missing, or more than a few chunks are noise; 1 = mostly generic or off-topic text.
- **Answer:** 3 = every part answered correctly from the context; 2 = partially correct (a part missing, or a minor misattribution); 1 = key parts missing (for example a comparison declined because one side is absent) or a fabricated claim.

As an objective cross-check I also computed **document coverage**: the fraction of the documents a question needs (Q1, Q2, Q5: one; Q4: two; Q3: three) that appear among the retrieved chunks.

2. Results and Analysis: Experiment A (embedding models)

Table 3. Experiment A, default parameters (chunk_size 1000, overlap 200, k 4). Averages over the five queries.

Model	Dim.	Avg retrieval (1-3)	Avg answer (1-3)	Doc coverage	Time to embed 1,065 chunks*
all-MiniLM-L6-v2	384	2.20	1.60	0.83	2.7 s
all-mpnet-base-v2	768	2.20	**2.00**	**0.93**	21.2 s

* Apple-silicon laptop (MPS), batch size 32, measured once after warm-up.

Table 4. Per-query scores (retrieval / answer) and the sources of the four retrieved chunks.

Q	MiniLM	MiniLM sources	mpnet	mpnet sources
Q1	2 / 2	Blueprint p.7, p.47, p.47, p.46	2 / 2	Blueprint p.7, p.8, p.8, p.2
Q2	3 / 2	EO p.7, p.7, p.7, p.13	2 / 2	EO p.7, p.13, p.13, p.7
Q3	2 / 1	NIST p.20, p.21, p.21; Blueprint p.14	2 / 2	NIST p.20, p.21; Blueprint p.9, p.14
Q4	2 / 1	NIST p.8, p.37, p.8, p.7	3 / 2	NIST p.47; EU p.21, p.26, p.15
Q5	2 / 2	EU p.27, p.14, p.10, p.46	2 / 2	EU p.46, p.78, p.80, p.14

The average retrieval scores tie, but the averages hide what matters. MiniLM's chunks were usually on-topic *within one document*: on both cross-document questions it filled its slots from a single document (NIST) and never reached the EU AI Act. mpnet sometimes pulled a weaker chunk, but on

Q4 it reached the second document, and on Q3 its second-document chunk carried a real duty instead of boilerplate. That is why its answer score is higher. Coverage shows the same direction: 0.83 vs 0.93.

2.1 Qualitative comparison, question by question (best = mpnet, worst = MiniLM)

Q1 (Blueprint, single document): same score, different failure.

MiniLM's four chunks were all about the right principle, but only two pieces of text were distinct. The principle statement is printed twice in the Blueprint (the summary on p. 7 and the principle page on p. 46), and both copies ranked in the top 4, along with two paragraphs of "Why this principle is important" from p. 47. More importantly, the 1000-character cut on p. 7 falls exactly after "...should not impose an unreasonable burden on the public." The very next sentence, "Automated systems with an intended use within sensitive domains, including ... criminal justice, employment, education, and health...", went into the following chunk, which did not rank. The generator handled this correctly: it answered the opt-out and fallback parts accurately and said that "the excerpts do not specify which sensitive domains call for extra human oversight." mpnet retrieved that second chunk, with the domain list, at rank 1, but lost the opt-out sentence. Its other three chunks were generic Blueprint framing (the "overlapping set of backstops" passage, the two-part scope test, and the legal disclaimer on p. 2). Its generator then stitched the pieces together wrongly: "people should be able to opt out of an automated system *when it is used in sensitive domains*". The Blueprint never says that. Both runs scored 2/2, but MiniLM's answer is incomplete and honest, while mpnet's is incomplete *and* contains a misattribution.

Q2 (Executive Order, single document): MiniLM slightly better at retrieval.

Both models found Sec. 4.2 on p. 7. MiniLM spent three of its four slots there: the reporting list (A)-(B), the threshold paragraph, and the "technical conditions" paragraph. mpnet spent two slots on Sec. 4.6 (p. 13, models with widely available weights). That section repeats the phrase "dual-use foundation models" six times but is about a different policy question. The answers were identical in content and correct on all three thresholds. The generator converted the flattened "10 26" and "10 20" into 10^{26} and 10^{20}, so it repaired a PDF artifact that would mislead a keyword system. Both answers missed reporting item (C), the results of red-team testing, because the default chunk ends at item (B) and the next chunk did not rank.

Q3 (multi-document synthesis): both models miss the EU Act; only mpnet's Blueprint chunk says anything.

Both models spent their best slots on the NIST section "3.4 Accountable and Transparent" (pp. 20-21). That is the right passage: it defines meaningful transparency, ties it to "actionable redress", and asks developers to "proportionally and proactively" adjust transparency when "life and liberty are at stake". MiniLM used three of its four slots there (p. 20 and both halves of p. 21). Its fourth chunk was a single Blueprint sentence saying the examples "are not critiques or endorsements" (p. 14), which carries no duty at all. Its answer covered NIST well, said that the Blueprint and the EU Act were not covered, and wrote a "common understanding" drawn from NIST alone (answer 1). mpnet used two

slots on NIST and two on the Blueprint. One of these was generic (p. 9), but the other (p. 14) states an actual Blueprint expectation: "Expectations about reporting are intended for the entity developing or using the automated system", with reports provided "to the public, regulators, auditors" and "made public as much as possible". Its answer therefore covered two of three documents with real duties, synthesized them, and flagged the EU Act as missing (answer 2).

The more important finding is what *neither* model retrieved. Ranking all 1,065 chunks against the query shows that the Blueprint's own "Notice and Explanation" principle ("You should know that an automated system is being used and understand how and why it contributes to outcomes that impact you", p. 6) ranked **774th** for MiniLM and **444th** for mpnet. Of the 23 Blueprint chunks that mention both "notice" and "explanation", the best-ranked was 194th for MiniLM and 191st for mpnet. Two forces push it down. First, the query says "transparency" while the Blueprint says "notice" and "explanation". Second, the query contains the long string "Blueprint for an AI Bill of Rights", which 30 Blueprint chunks repeat verbatim, mostly in introductions and boilerplate, so those chunks win on surface overlap. The EU articles were closer for mpnet, which is the clearest model difference on this question: Art. 13 ranked 6th (just outside k = 4) and Art. 52 22nd, against 82nd and 151st for MiniLM.

Q4 (comparative): MiniLM failed to retrieve a required document.

This is the clearest case of a model failing to retrieve a crucial document. All four MiniLM chunks came from NIST: the RMF overview on p. 8 (twice), the MANAGE table on p. 37, and the outline on p. 7. 44 of the 148 NIST chunks begin with the running header "NIST AI 100-1 AI RMF 1.0", so any query that names "NIST AI RMF" gets a lexical boost from text that carries no content. With no EU text, the generator declined the comparison ("The context provided does not include any information about the EU AI Act"), which is the correct behavior but scores 1. mpnet retrieved NIST Appendix D (p. 47, the RMF should "be risk-based, resource-efficient, pro-innovation, and voluntary") plus three EU chunks, including p. 15 on the "ex-ante and ex-post compliance and enforcement mechanisms" and third-party conformity assessment. It produced a clean three-axis comparison. It still scored 2, not 3: for "who must comply" under the EU Act, it relied on recital 12 (p. 21) and wrote that the regulation "applies to Union institutions, offices, bodies, and agencies". That recital is about extending the regulation to EU institutions, not about who is primarily bound (providers and users).

Q5 (definition and scope): mpnet finds the article, MiniLM finds paraphrases of it.

Neither model retrieved the Art. 3(1) definition on p. 40, so both answers correctly said the definition was not in the context. (In the pilot, when the question said "artificial intelligence system" instead of "AI system", mpnet did retrieve Art. 3(1) at rank 1. The shorter wording plus "Article 6" pulled every top result toward classification.) The difference is in the Article 6 half. mpnet's rank-1 chunk is the actual text of Art. 6(1) with both conditions (a) and (b), and its answer quotes them exactly. MiniLM's chunks were recital 32 (p. 27), the explanatory memorandum (p. 14), the policy-options page (p. 10) and the *second* half of p. 46 (Art. 6(2) and Art. 7). Its answer says that under Article 6 a system is high-risk "if, based on its intended purpose, it poses a high risk of harm to health and safety or fundamental rights ... severity ... probability". That wording comes from the recital and from Art. 7 (the criteria for *amending* Annex III), not from Art. 6. Both runs scored 2/2, but mpnet's answer is

precisely correct for the half it answers, while MiniLM's is a plausible paraphrase of the wrong provision.

2.2 Correlation between retrieval quality and answer quality

Across all 55 scored runs (both experiments), the Pearson correlation between retrieval and answer scores is **0.61**. More telling, the answer score **never exceeded** the retrieval score (55 of 55 runs have answer <= retrieval). The joint counts were (retrieval, answer) = (2,2) x 23, (2,1) x 17, (3,2) x 7, (1,1) x 4, (3,3) x 4. Good retrieval is necessary but not sufficient: the 24 runs that lost a point between retrieval and answer are the interesting ones.

Generator failures (good retrieval, weaker answer):

- **Chunk size 2000, Q5.** The context contained Art. 6(1) and Art. 6(2) ("AI systems referred to in Annex III shall also be considered high-risk"). The answer reported only the two conditions of paragraph 1 and silently dropped paragraph 2, the second route to high-risk status.
- **Overlap 400, Q1.** The domain list was not in the context. Instead of saying so, as the default run did, the model invented an answer: "sensitive domains ... include situations where rights, opportunities, or access are meaningfully at stake, as well as cases where the automated system may reinforce bias". That is a rewording of the "why this matters" paragraph presented as the requested list.
- **k = 2 and k = 12, Q5.** Despite the instruction to use only the context, both runs began "The EU AI Act proposal defines an 'AI system' as a system that is capable of performing tasks that typically require human intelligence" (k = 2), or "...any system that uses artificial intelligence techniques..." (k = 12), and then admitted that the definition was not in the excerpts. These are textbook definitions from the model's pre-training, not the Act's text (which says "software that is developed with one or more of the techniques and approaches listed in Annex I..."). Both received answer score 1.
- **k = 8, Q3: a chunk repurposed for the wrong question.** The context included the Blueprint's human-fallback text (p. 7), which is about opt-out and human review, not transparency. The answer cited it as the Blueprint's transparency duty: the Blueprint "emphasizes the need for transparency in automated systems, particularly in sensitive domains such as criminal justice, employment, education, and health". The Blueprint does not say that, so the run scored 1. The k = 12 run repeated the same misattribution.
- **Citation errors.** With k = 8 on Q1, the model cited the p. 7 chunk as "Excerpt 5, p. 49". The fact was right but the provenance was wrong, which matters in a policy setting.

Generator successes (imperfect retrieval, good answer):

- **Q2, all runs.** The threshold chunk is corrupted by flattened superscripts ("greater than 10 26 integer or floating-point operations", "greater than 1023"). gpt-4o-mini rendered all three thresholds correctly as 10^{26}, 10^{23} and 10^{20}.
- **k = 12, Q4.** The only EU chunk was recital 32, at rank 11 of 12, below ten NIST chunks. The generator still found it and built a correct three-axis comparison. It combined NIST p. 13

("development and deployment should cease in a safe manner until risks can be sufficiently managed") and p. 12 ("does not prescribe risk tolerance") with the recital's third-party conformity assessment. One relevant chunk at the bottom of a long context was enough.

- **Chunk size 500, Q3.** Retrieval was only partially relevant (retrieval 2): no Blueprint duty and a single EU chunk. But that chunk was the opening sentence of Art. 52 ("Providers shall ensure that AI systems intended to interact with natural persons are designed and developed in such a way that natural persons are informed that they are interacting with an AI system"). The generator turned it into a precise EU duty, paired it with NIST's "meaningful transparency", and stated plainly that the Blueprint excerpt "does not explicitly outline specific transparency and explanation duties" (answer 2).

2.3 Which model performed best, and why

all-mpnet-base-v2 performed best overall on this corpus: higher answer score (2.00 vs 1.60), higher document coverage (0.93 vs 0.83), and it was the only model to reach the EU AI Act on the comparative question (Q4), to retrieve a substantive Blueprint duty on Q3, and to retrieve the exact Article 6 text on Q5. Even where it missed, it missed by less: EU Art. 13 ranked 6th for mpnet on Q3 and 82nd for MiniLM. MiniLM won only Q2, a single-document question with strong lexical overlap between query and source.

I think three factors explain this. (1) **Capacity for paraphrase.** mpnet has twice the layers and twice the embedding width. The questions most often describe a concept in words different from the source ("what happens before a system can be deployed" vs "ex-ante conformity assessment"; "transparency duties" vs "Transparency and provision of information to users"), and that is where a larger encoder helps. (2) **MiniLM is more swayed by surface strings.** Both models suffered from it on Q3, but MiniLM's retrievals were more often dominated by chunks that repeat the document *names* in the query ("Blueprint for an AI Bill of Rights", "NIST AI 100-1 AI RMF 1.0" headers). In a corpus where three of four documents name themselves constantly and the EU proposal calls itself "this Regulation" (only 2 of 424 EU chunks contain "Artificial Intelligence Act"), that bias systematically starves the EU document. (3) **Longer context window.** mpnet reads 384 tokens versus MiniLM's 256, which matters little at chunk size 1000 but a lot for longer chunks (Section 3). The price is about 8x slower embedding (21.2 s vs 2.7 s for this corpus) and twice the storage per vector. That is negligible here, but it would matter for millions of chunks or for embedding queries on a CPU at serving time.

3. Results and Analysis: Experiment B (chunk size, overlap, k with MiniLM)

Design. Starting from the default configuration (1000 / 200 / 4, the MiniLM run of Experiment A), I changed one parameter at a time and re-ran all five queries, rebuilding the index for every chunking change. Values: chunk_size in {250, 500, 1000, 2000} (overlap 200, lowered to 50 for size 250 because a 200-character overlap on a 250-character chunk would make neighbours 80% identical); chunk_overlap in {0, 100, 200, 400} (size 1000); k in {2, 4, 8, 12}. Every one of the 45 runs was scored with the same rubric.

Table 5. Experiment B results (all-MiniLM-L6-v2). Per-query entries are retrieval / answer.

Configuration	Chunks	Q1	Q2	Q3	Q4	Q5	Avg ret.	Avg ans.	Cov.
size 250 (overlap 50)	3,922	2/1	2/1	2/1	1/1	1/1	1.60	1.00	0.90
size 500	2,334	2/2	2/2	2/2	2/1	1/1	1.80	1.60	0.90
default (1000 / 200 / 4)	1,065	2/2	3/2	2/1	2/1	2/2	2.20	1.60	0.83
size 2000	546	3/3	3/3	1/1	2/1	2/2	2.20	2.00	0.83
overlap 0	1,009	2/2	3/2	2/2	2/1	2/2	2.20	1.80	0.83
overlap 100	1,014	2/2	3/2	2/2	2/1	2/2	2.20	1.80	0.83
overlap 400	1,221	2/2	3/2	2/1	2/1	2/2	2.20	1.60	0.83
k = 2	1,065	2/2	3/2	2/1	2/1	2/1	2.20	1.40	0.77
k = 8	1,065	3/3	3/2	2/1	2/1	2/2	2.40	1.80	0.83
k = 12	1,065	3/3	2/2	2/2	2/2	2/1	2.20	2.00	1.00

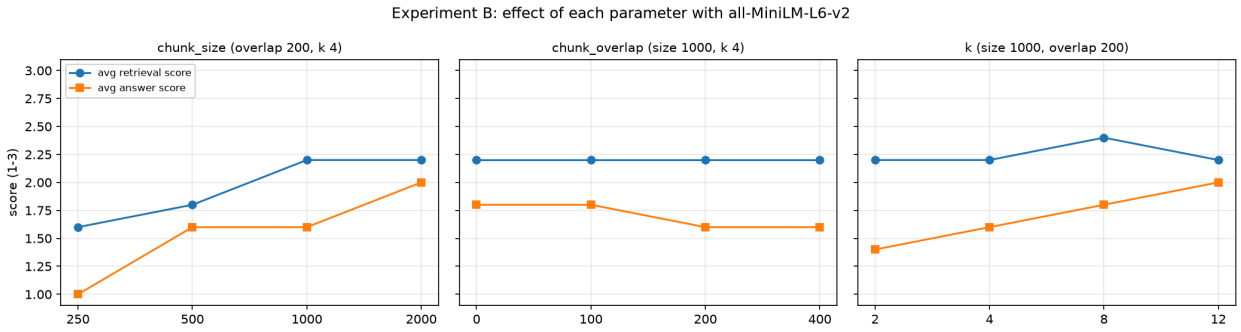


Figure 1. Average retrieval and answer score as each parameter varies (others at default).

Chunk size had the largest effect, in both directions.

Small chunks (250, 500) hurt retrieval (1.60 and 1.80 vs 2.20). Size 250 also gave the worst answers (1.00); size 500 tied the default on answers (1.60) only because of one telling exception on Q3, described below. At 250 characters a chunk is one or two sentences (median 38 tokens). For Q5, it returned four fragments of procedural text that happen to contain "high-risk AI systems" ("For high-risk AI systems referred to in point 1(a) of Annex III, the measures referred to in paragraph 3 shall be such as to ensure that, in addition,"). Short chunks make the vector a fingerprint of a few words, so retrieval matches surface phrases, and even a correct hit carries too little context to answer from. On Q2, the size-250 retrieval contained only the 10^{26} part of the threshold sentence, and the answer reported that the cluster threshold was "not detailed in the excerpts". Average answer score: 1.00. The exception is Q3, where small chunks were the *only* chunk sizes that reached EU Article 52 (document coverage rose to 0.90). A short chunk that begins with a heading is dominated by that heading, and the heading "TRANSPARENCY OBLIGATIONS FOR CERTAIN AI SYSTEMS" matches the query almost perfectly. At size 250, though, the retrieved fragment was *only* the heading ("Article 52 Transparency obligations for certain AI systems 1."), so the answer could say nothing more than that such obligations exist (answer 1). At size 500 the fragment also contained the first sentence of the

duty, which was enough for a correct EU answer (answer 2). Small chunks raise the chance of hitting a heading and lower the chance that the hit contains anything usable.

Large chunks (2000) gave the best answers on single-document questions (Q1 and Q2 both 3/3). This was the only configuration in which a whole provision fit into one retrieved chunk. The full Blueprint principle (opt-out, fallback, *and* the domain list) and the whole of EO Sec. 4.2(a), including red-team item (C), arrived intact, so the generator had nothing to miss. But Q3 dropped to retrieval 1. A 2000-character chunk usually spans several topics, so its embedding is a blurred average, and three generic, title-heavy Blueprint chunks (p. 4 "About this framework", the p. 14 companion introduction, the p. 2 disclaimer) took three of the four slots, leaving only one NIST transparency chunk. A second, hidden mechanism makes this worse: **65.8% of the 2000-character chunks exceed MiniLM's 256-token input limit** (and 9.9% exceed mpnet's 384), so the embedder silently truncates them. The retriever ranks a chunk by roughly its first 1100 characters, while the generator receives all 2000. That explains the pattern in Table 5: chunk size 2000 improved the answers much more than the retrieval.

Chunk overlap had a small and erratic effect.

Q1, Q2, Q4 and Q5 scored identically under all four overlap settings. Only Q3 moved: overlap 0 and 100 scored 2/2, while the default 200 and 400 scored 2/1, so the averages are 1.80, 1.80, 1.60 and 1.60. The difference is one chunk. At overlap 0 and 100, the explanatory-memorandum passage on EU p. 4 ("For some specific AI systems, only minimum transparency obligations are proposed, in particular when chatbots or 'deep fakes' are used") entered the top 4. At 200 and 400, a generic Blueprint chunk took that slot. That is not a trend: more overlap did not steadily help or hurt. Changing the overlap moves every chunk boundary, which slightly changes every embedding and reshuffles the order near the cut-off at rank 4. On the other queries the chunks also shifted (up to three of the four per query, usually to a neighbouring passage of the same section), but without changing the evidence available. Structurally, splitting is per page and prefers sentence boundaries, so most cuts already fall between sentences, and the overlap only copies whole trailing sentences into the next chunk. More overlap also adds near-duplicates (1,221 chunks at overlap 400 vs 1,009 at 0) that compete for the same top-k slots. Overlap is cheap insurance against a fact being split mid-sentence, but it did not rescue the Q1 domain list, which sits in the *next* chunk and never ranks. The practical lesson is that results near the rank-k cut-off are fragile: one score point can come from which chunk happens to land at rank 4.

k traded precision for recall, and the generator benefited from recall.

The answer score rose monotonically with k: 1.40 (k = 2), 1.60 (4), 1.80 (8), 2.00 (12). Retrieval relevance peaked at k = 8 (2.40) and fell at k = 12. For Q2, k = 12 added five clearly irrelevant chunks (an EU article on small-scale providers, the Blueprint's national-security disclaimer, an EO workforce section), so the retrieved set was no longer "highly relevant". The gains came from the long tail. k = 8 pulled in the second half of the Blueprint p. 7 principle at rank 5, finally answering Q1 in full. k = 12 was the **only** MiniLM configuration that reached the EU AI Act on Q4, and the only one with full document coverage (1.00). On Q4, a single EU chunk at rank 11 was enough to turn a declined comparison into a 2/2 answer. On Q3, two EU summary passages at ranks 10 and 12 (p. 7 and p. 8, "only very limited transparency obligations are imposed" for non-high-risk systems) gave the answer

an EU section. k = 2 was too few: every multi-part question lost a part. More context is not free, though. k = 12 (and k = 2) produced the fabricated "AI system" definition on Q5, and the k = 12 prompts were about three times longer, which costs latency and tokens. For a model like gpt-4o-mini, which handled 12 excerpts well, k around 8 gave the best balance, together with deduplication: identical principle text at p. 7 and p. 46 wasted a slot in almost every Q1 run.

Summary of Experiment B. Chunk size changes *what an embedding represents* and had the largest effect. Too small loses context; too large blurs the vector and gets truncated by the encoder. k changes *how much evidence the generator sees*; a larger k raises answer quality and document coverage at the cost of precision, prompt length and some hallucination risk. Overlap had the smallest effect, and that effect was erratic: it only reshuffled which chunk landed at rank 4 on Q3. No parameter setting fixed the core weakness of MiniLM on this corpus, its inability to reach the EU AI Act on the comparative question, except brute-force k = 12. No setting at all retrieved the Blueprint's own Notice and Explanation principle for Q3.

4. Conclusion

The single most important lesson: in a multi-document corpus, recall across documents, not the LLM, is the bottleneck, and it has to be engineered and checked deliberately. In 55 of 55 runs the answer was no better than the retrieved evidence. The generator was good at reading broken text (10^{26} from "10 26"), at using a single relevant chunk buried at rank 11, and at admitting gaps. But it cannot synthesize a document that never reaches its context, and with thin context it sometimes filled gaps from its own pre-training. Every major failure in this study was a retrieval failure: MiniLM never reaching the EU Act on Q4, the Blueprint's Notice and Explanation principle ranking 444th or lower for both models on Q3, the domain sentence just past a chunk boundary, and the duplicated principle text taking up slots. Document-name strings and running headers bias similarity toward documents that repeat their own names. Chunk size and the embedder's token limit interact in ways invisible unless you look at the chunks.

In practice, I would (1) choose the stronger embedder (mpnet here) and keep chunks within its token limit; (2) make cross-document coverage explicit, either by decomposing a comparative or synthesis question into one sub-query per named document, or by metadata-filtered retrieval of the top few chunks *per document*, instead of hoping one global top-k covers all sources; (3) deduplicate near-identical chunks or use maximal marginal relevance, and strip running headers before embedding; and (4) evaluate by reading the retrieved chunks, as this assignment required. The averages in Table 3 tie on retrieval; only reading the chunks showed that one model was missing an entire document on Q4, and that both models missed the most relevant Blueprint principle on Q3.

Limitations. Five queries and a single human grader on a coarse 3-point scale; scores are judgments, and a different grader could move individual cells by one point. Answers come from a single temperature-0 sample per configuration (cached for reproducibility). Experiment B was run only for MiniLM, as specified.